

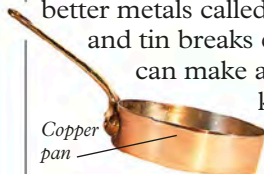


METALS



Iron ore

Metals are found in the ground, hidden in certain rocks called ores. Tin, copper, and iron all have to be taken out of their ores before a factory can melt and shape them into a can, pan, or car. Pure metals, however, are usually too weak to be used in industry, and have to be mixed together to make better metals called alloys. Lead is soft, and tin breaks easily. Together they can make a strong, tough alloy known as pewter.



Copper pan

Not natural

Brass cannot be dug out of the ground. It is an alloy made by mixing together two weaker metals, copper and zinc.

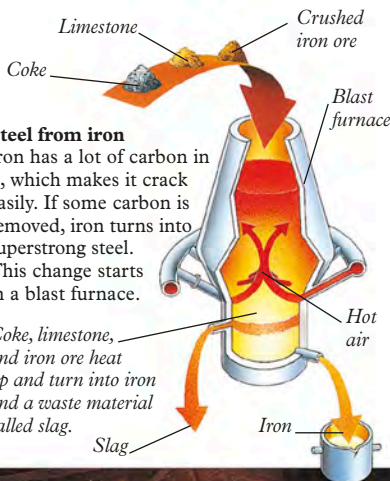
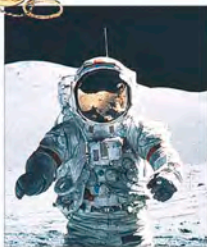
Brass is stronger than zinc or copper.



Zinc-coated bucket and wire

Precious metals

Gold and silver are used to make much more than just jewelry. Gold is sprayed onto an astronaut's visor to reflect sunlight. Silver is used in electronic equipment because it carries electricity very well.



Steel from iron

Iron has a lot of carbon in it, which makes it crack easily. If some carbon is removed, iron turns into superstrong steel. This change starts in a blast furnace.

Coke, limestone, and iron ore heat up and turn into iron and a waste material called slag.



1. Blasted iron ore

A stream of iron is flowing from this huge oven, called a blast furnace. It has been burned out of iron ore by blasts of hot air.

